STM32H750 Flight Controller Prototype

Project Description

The **STM32H750 Flight Controller Prototype** is a high-performance embedded system designed as the foundation for developing custom unmanned aerial vehicle (UAV) and drone flight control systems. Built around the **STM32H750VBT7** microcontroller featuring a **480 MHz ARM Cortex-M7 processor**, the controller provides the computational power required for real-time sensor fusion, flight stabilization, and motor control.

The prototype interfaces with essential flight sensors—including a **6-axis Inertial Measurement Unit (IMU)**, **3-axis magnetometer**, and **high-precision barometric pressure sensor**—using the **SPI communication protocol** for high-speed and reliable data transfer. These sensors provide accurate measurements of acceleration, angular velocity, heading, and altitude, enabling precise flight control and navigation.

The controller generates multiple **PWM outputs** for driving Electronic Speed Controllers (ESCs) and brushless motors while also providing **I^2C**, **SPI**, and **GPIO interfaces** for future expansion with GPS modules, telemetry radios, additional sensors, and other flight peripherals.

Designed with a custom PCB layout, the prototype emphasizes reliable power distribution, signal integrity, and modular hardware architecture, making it suitable for research, educational projects, and the development of advanced autonomous drone systems.

Hardware Components

Component	Quantity	Purpose
STM32H750VBT7 Microcontroller	1	Main flight controller and real-time processing unit
16 MHz Crystal Oscillator	1	System clock source
10 pF Capacitors	2	Crystal oscillator load capacitors
ICM-20602 IMU	1	3-axis accelerometer and 3-axis gyroscope for motion sensing
LIS3MDL Magnetometer	1	Electronic compass for heading estimation
MS560702BA03-50 Barometer	1	Atmospheric pressure sensor for altitude measurement
SPI Communication Bus	Multiple	High-speed communication with onboard sensors
PWM Output Channels	4	Motor and ESC control signals
I^2C Interface	1	Expansion interface for external peripherals
GPIO Interfaces	Multiple	General-purpose input/output for auxiliary devices

Component	Quantity	Purpose
Power Input (VIN)	1	Main power supply input
Ground Connections	Multiple	Stable electrical grounding
Custom PCB	1	Compact hardware integration and signal routing

Key Features

- High-performance STM32H750 ARM Cortex-M7 microcontroller
- 480 MHz processing for real-time flight control
- SPI-based communication with all onboard sensors
- Integrated IMU, magnetometer, and barometric pressure sensor
- Four PWM outputs for ESC and brushless motor control
- I^2C, SPI, and GPIO expansion interfaces
- Custom PCB optimized for embedded flight systems
- Low-latency sensor data acquisition
- Modular hardware architecture for future upgrades
- Suitable for autonomous UAV and drone development

Applications

- Multirotor drone flight controllers
- Autonomous UAV research
- Educational embedded systems projects
- Flight control algorithm development
- Robotics and navigation systems
- Sensor fusion and attitude estimation
- Custom autopilot development
- Aerospace and embedded electronics prototyping